Titanic Dataset Exploratory Data Analysis (EDA)

Internship Task 5 – Elevate Labs

Prepared By

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1. Introduction

Exploratory Data Analysis (EDA) is a crucial step in the data analysis process that helps in understanding the structure, characteristics, and patterns present within a dataset. The objective of this project is to perform EDA on the Titanic dataset using Python libraries such as Pandas, Matplotlib, and Seaborn.

The analysis focuses on identifying missing values, understanding feature distributions, examining relationships between variables, and discovering the factors that influenced passenger survival.

2. Dataset Overview

The Titanic dataset contains information about passengers who travelled aboard the Titanic. The dataset includes demographic information, ticket details, travel class, fare information, and survival status.

Dataset Information

- Total Records: 891
- Total Features: 12

Features

- PassengerId
- Survived
- Pclass
- Name
- Sex
- Age
- SibSp
- Parch
- Ticket
- Fare
- Cabin

- Embarked

Target Variable

Survived

- 0 = Passenger Did Not Survive
- 1 = Passenger Survived

Dataset Shape

- Rows: 891
- Columns: 12

3. Data Quality Assessment

The dataset was examined for missing values before performing analysis.

Missing Values Summary

Column	Missing Values
Age	177
Cabin	687
Embarked	2

Observations

- Age contains several missing values.
- Cabin contains a significant number of missing values.
- Embarked contains only a few missing values.
- Remaining columns contain complete information.

4. Data Cleaning

To improve data quality and ensure accurate analysis, the following preprocessing steps were performed:

Data Cleaning Steps

1. **Missing values in the Age column were replaced using the median value.**
2. **Missing values in the Embarked column were replaced using the mode value.**
3. **The Cabin column was removed due to excessive missing values.**

Benefits

- **Improved dataset completeness.**
- **Reduced impact of missing data.**
- **Enhanced reliability of statistical analysis and visualizations.**

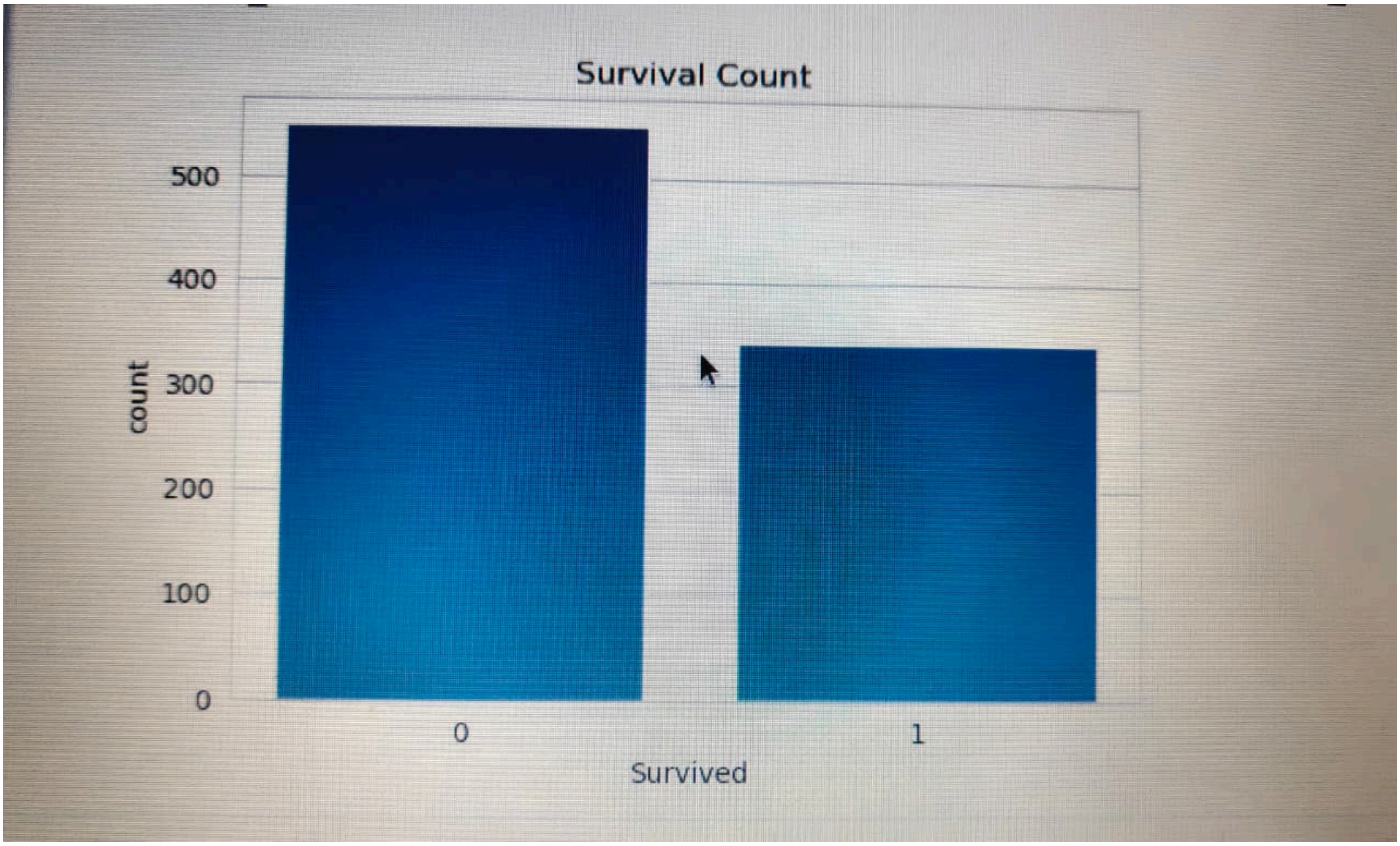
5. Univariate Analysis

Survival Distribution

A count plot was generated to analyze the distribution of passenger survival.

Observation

- **A larger proportion of passengers did not survive.**
- **The overall survival rate was lower than the non-survival rate.**

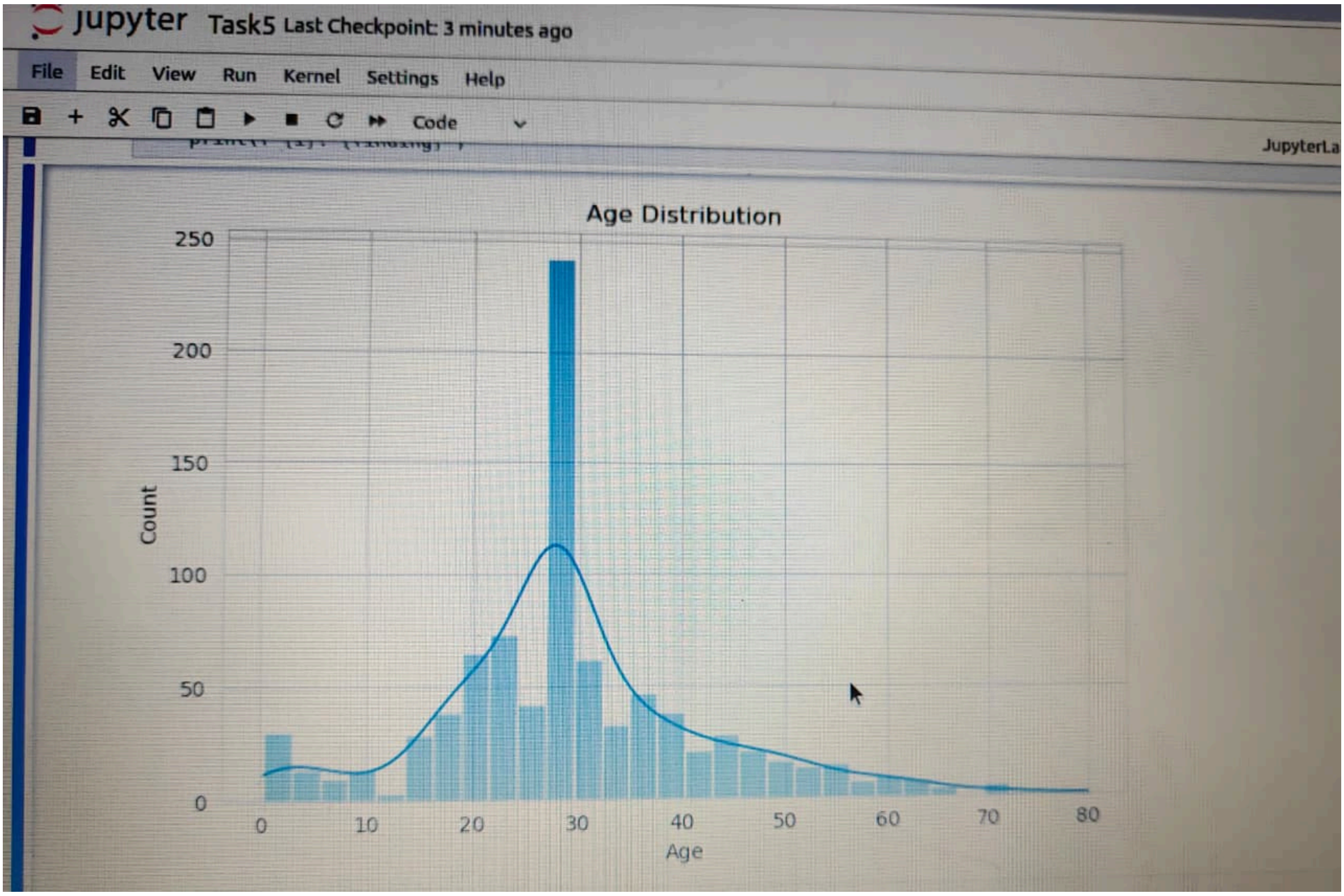


Age Distribution

The age distribution was visualized using a histogram with a density curve.

Observation

- **Most passengers belonged to the young adult age group.**
- **Passenger ages ranged from infants to elderly individuals.**
- **The average passenger age was approximately 29.7 years.**



Fare Distribution

A box plot was used to examine fare values and identify outliers.

Observation

- **Fare distribution is highly skewed.**
- **Several extreme outliers exist.**
- **A small number of passengers paid significantly higher fares.**

6. Bivariate Analysis

Survival by Gender

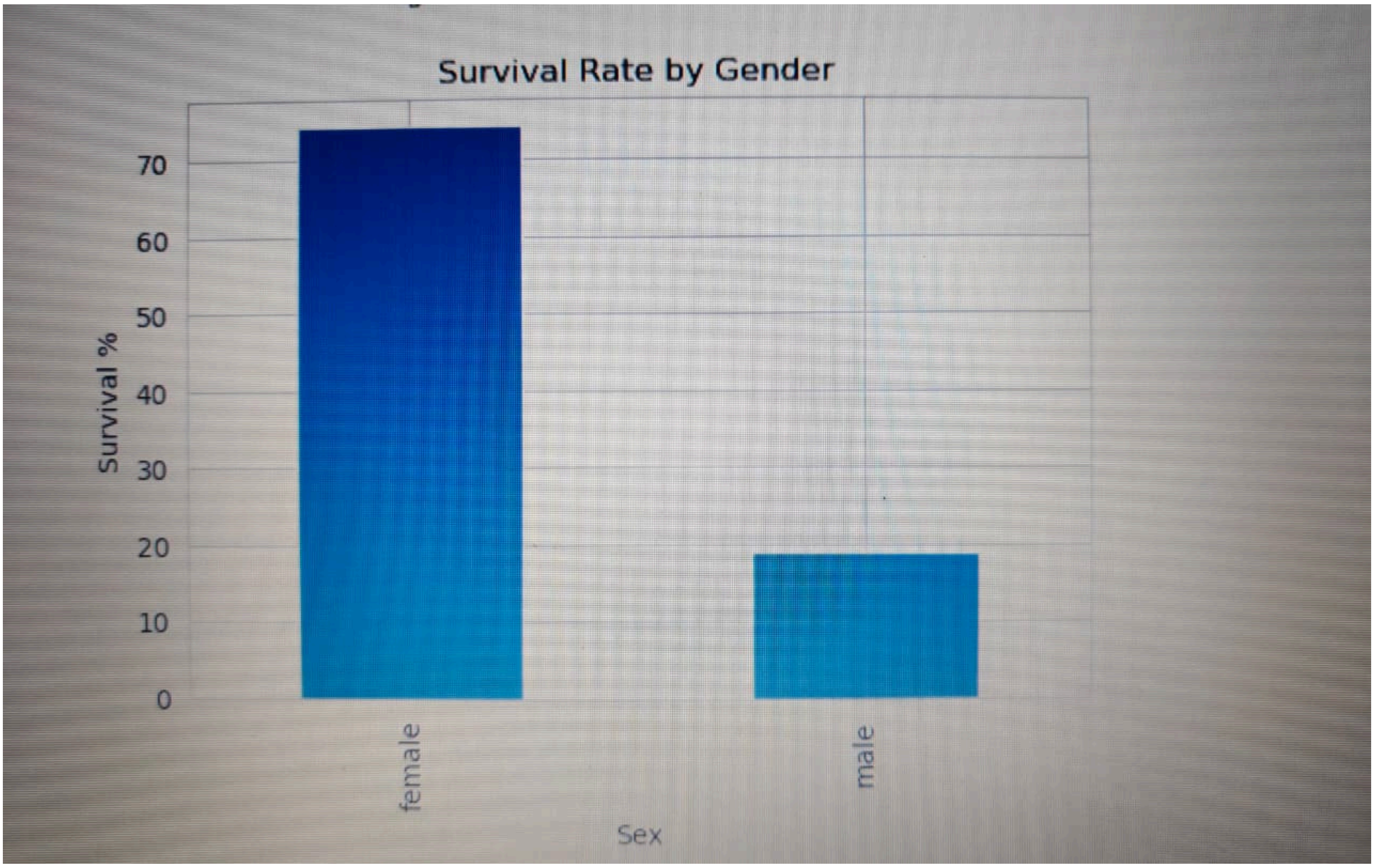
Passenger survival was analyzed based on gender.

Results

Gender	Survival Rate
Female	74.20%
Male	18.89%

Observation

Female passengers had a significantly higher survival rate compared to male passengers, suggesting that women received higher priority during evacuation.



Survival by Passenger Class

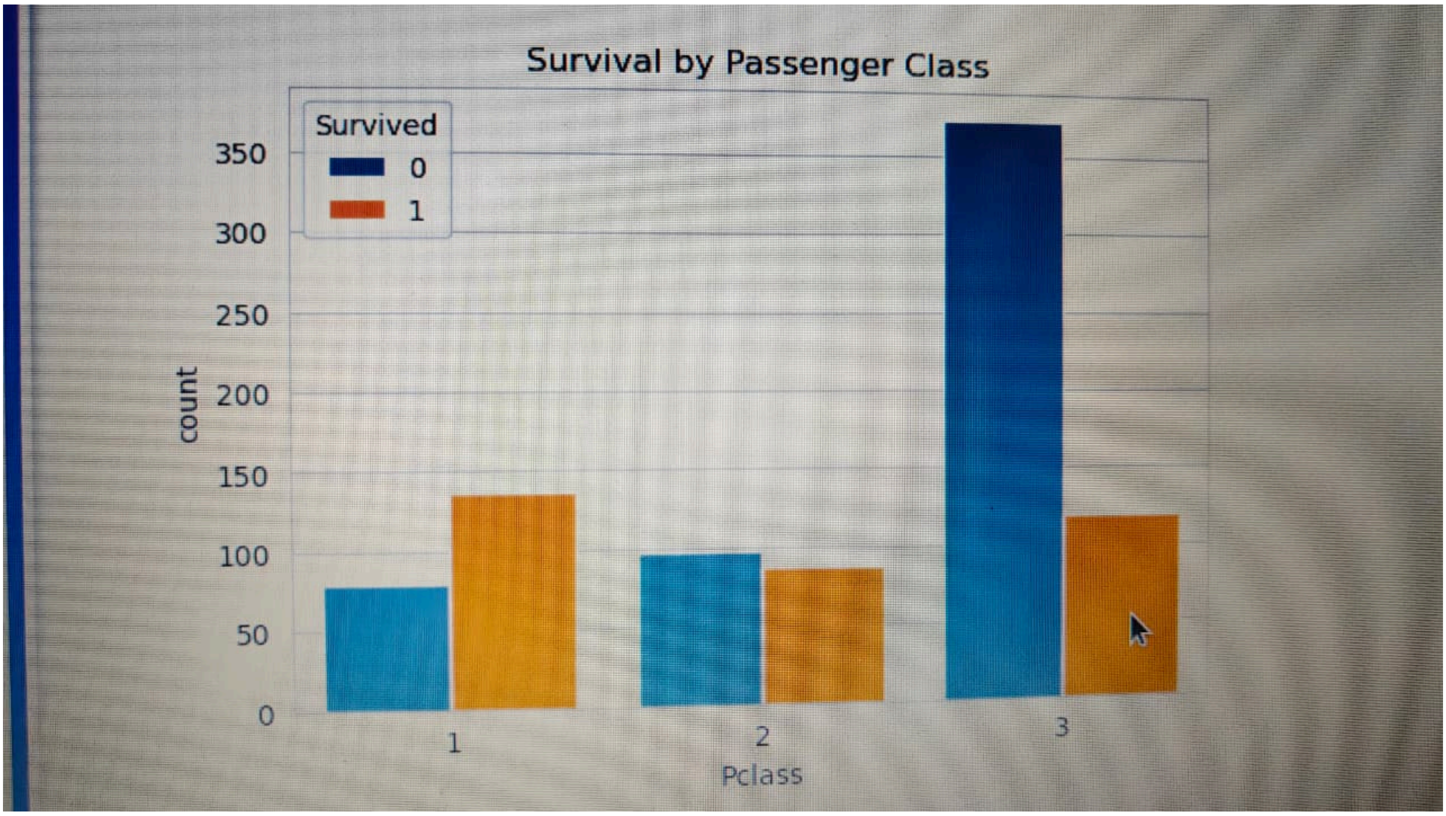
Passenger class was analyzed to determine its impact on survival.

Results

Passenger Class	Survival Rate
First Class	62.96%
Second Class	47.28%
Third Class	24.24%

Observation

Passengers traveling in First Class had the highest survival probability, while Third Class passengers had the lowest.

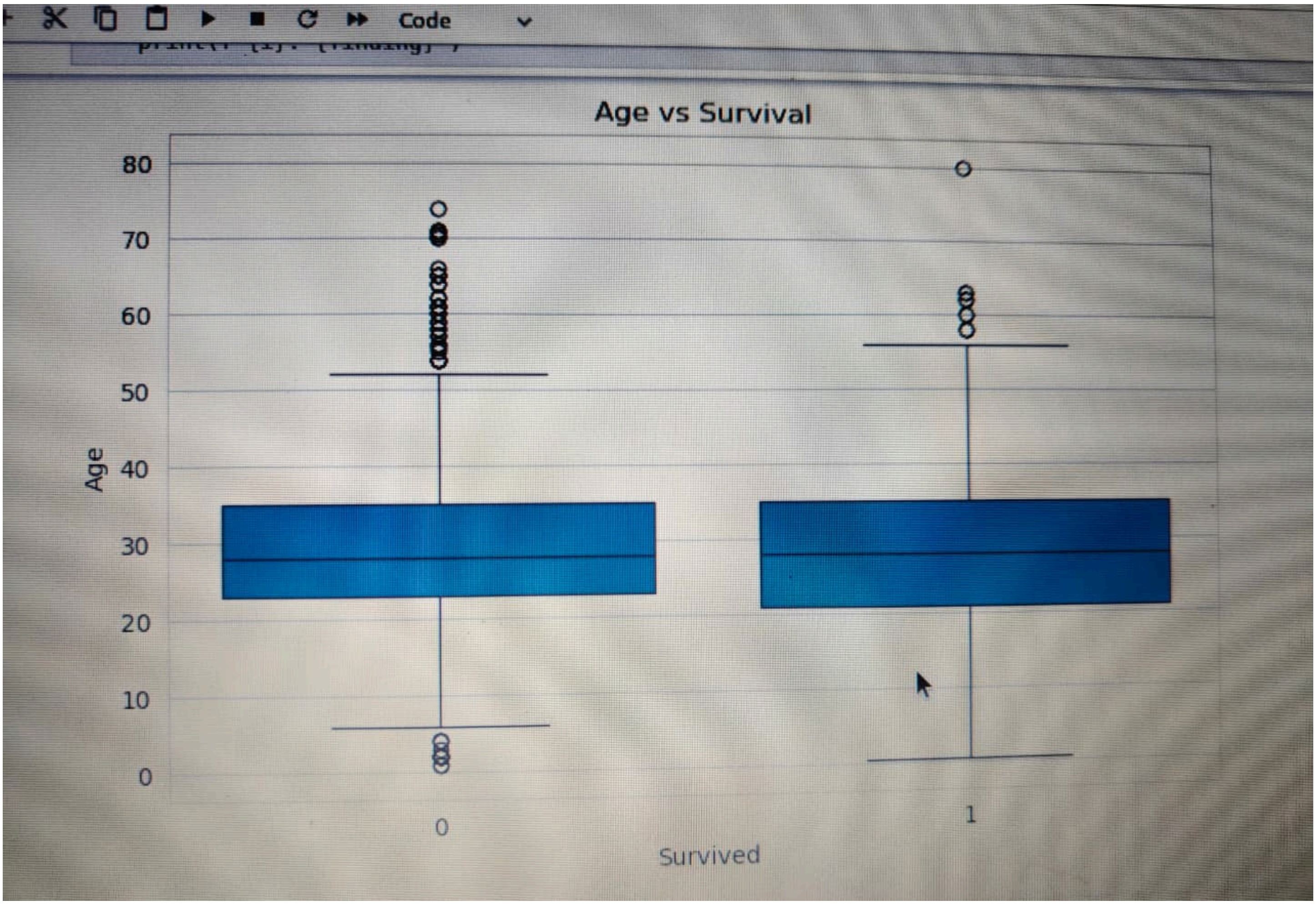


Age vs Survival

A box plot was used to compare age distributions among survivors and non-survivors.

Observation

- Younger passengers showed slightly better survival outcomes.
- Age had a moderate influence on survival compared to gender and passenger class.



7. Correlation Analysis

A correlation heatmap was generated to identify relationships among numerical variables.

Observation

- Gender showed a strong relationship with survival.
- Passenger class was strongly associated with survival outcomes.
- Fare demonstrated a moderate positive relationship with survival.
- Age showed a weaker correlation with survival.

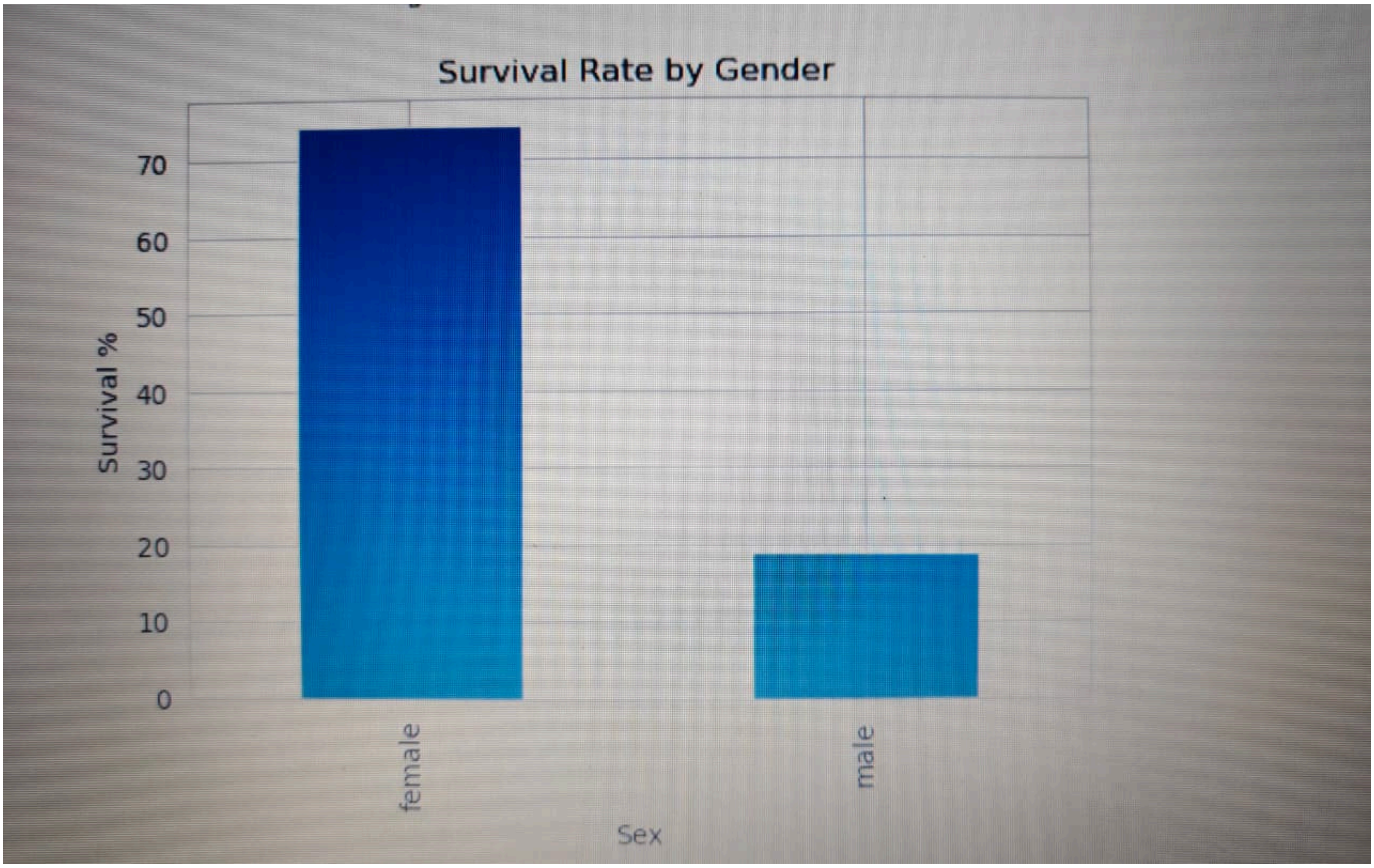
8. Additional Analysis

Survival Rate by Gender

The analysis revealed a significant difference in survival rates between male and female passengers.

Key Insight

Female passengers had a survival rate of 74.20%, while male passengers had a survival rate of only 18.89%.

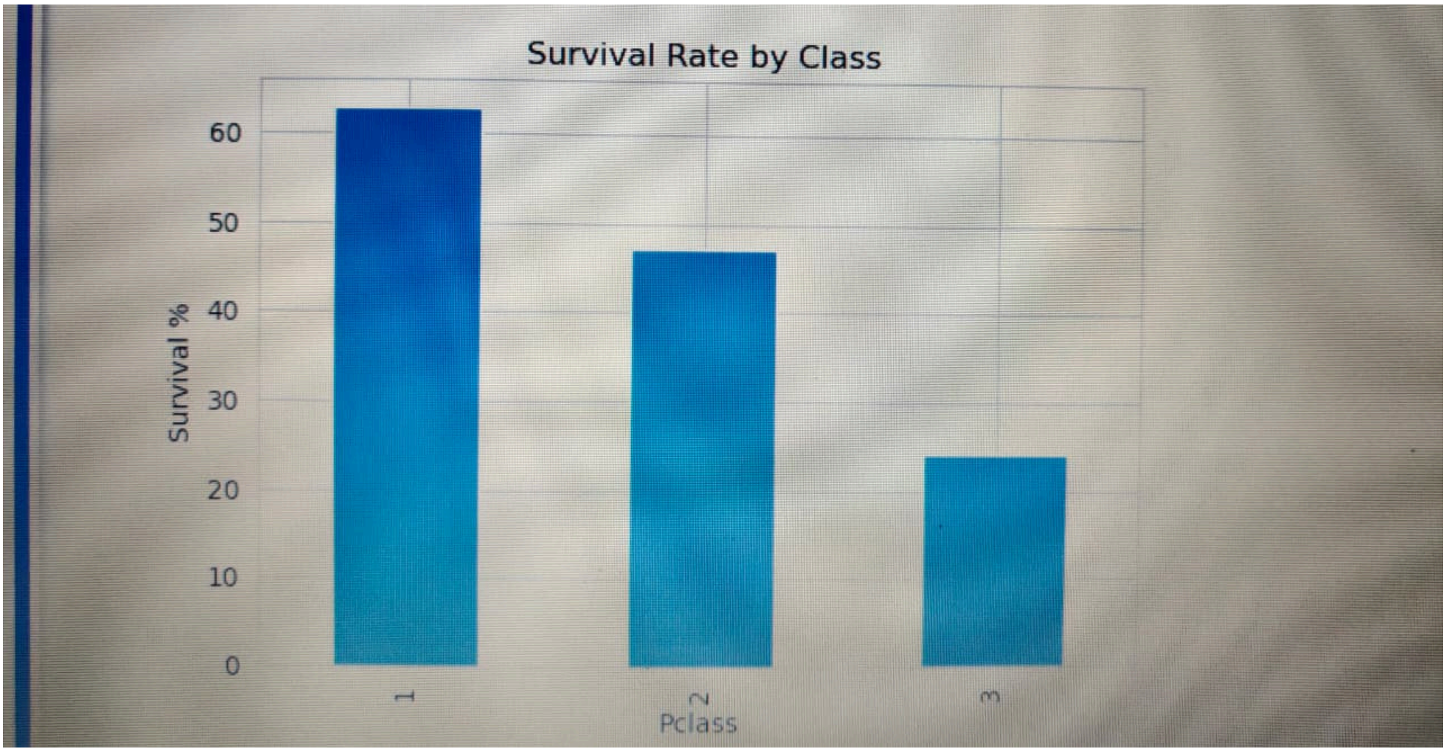


Survival Rate by Passenger Class

Passenger class played an important role in determining survival chances.

Key Insight

First Class passengers experienced the highest survival rates, highlighting the impact of socioeconomic status on survival.



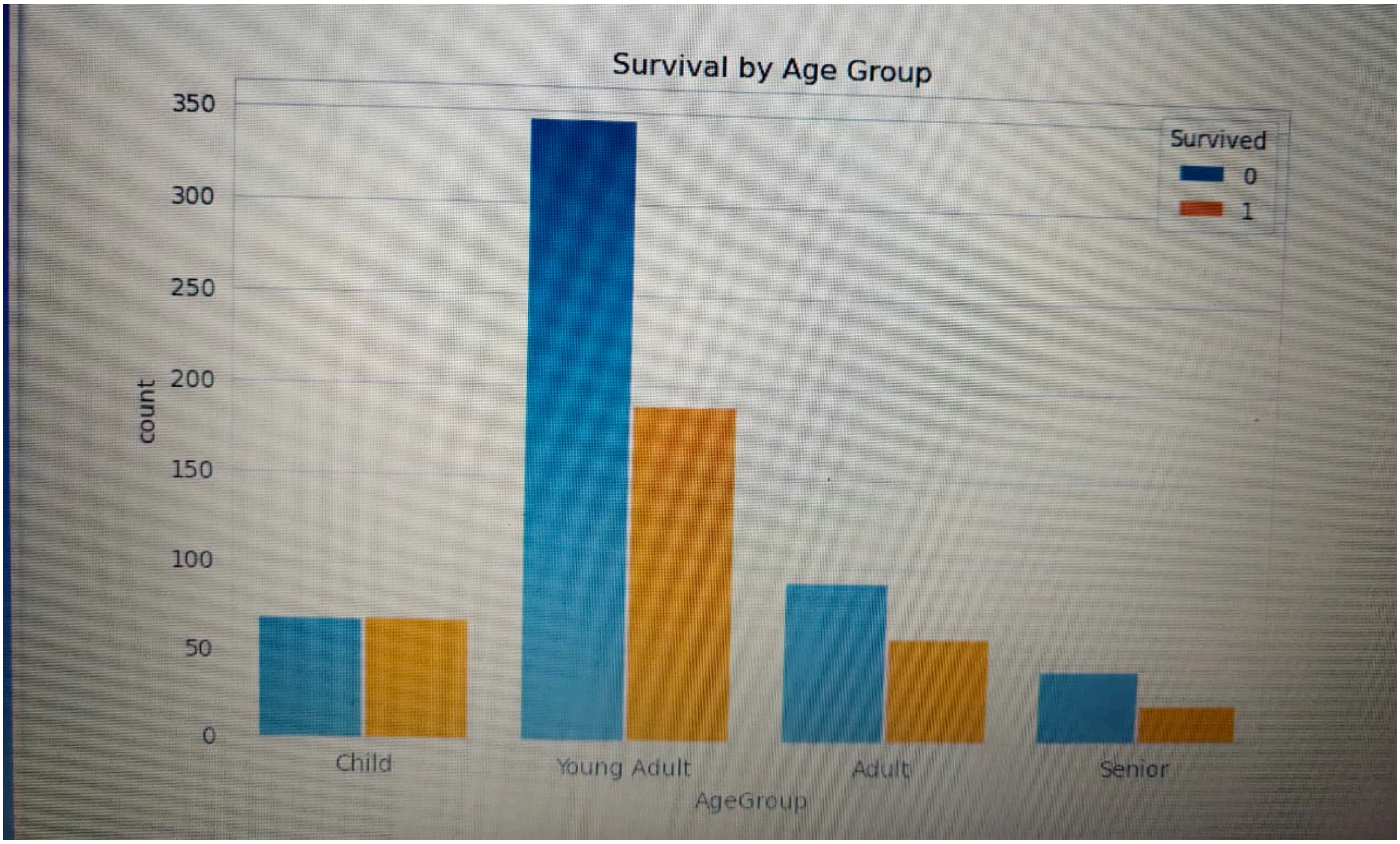
Age Group Analysis

Passengers were categorized into four age groups:

- **Child**
- **Young Adult**
- **Adult**
- **Senior**

Observation

- **Young adults formed the largest passenger group.**
- **Children generally experienced better survival outcomes.**



9. Key Findings

1. **Female passengers had a significantly higher survival rate (74.20%) than male passengers (18.89%).**
2. **Passenger class strongly influenced survival probability.**
3. **First Class passengers had the highest survival rate (62.96%).**
4. **Third Class passengers had the lowest survival rate (24.24%).**
5. **Most passengers were young adults between 20 and 40 years of age.**
6. **Fare distribution contained several high-value outliers.**
7. **Gender and Passenger Class emerged as the strongest predictors of survival.**
8. **Higher-paying passengers generally experienced better survival outcomes.**

10. Conclusion

The Exploratory Data Analysis of the Titanic dataset successfully identified several important factors that influenced passenger survival. Gender emerged as the most influential variable, followed by passenger class and fare. Female passengers and First Class travelers demonstrated significantly higher survival rates compared to other groups.

The analysis also highlighted the importance of data preprocessing, visualization, and statistical exploration in understanding real-world datasets. Through various visualizations and correlation analysis, meaningful insights were extracted that explain survival patterns among Titanic passengers.

This project demonstrates the practical application of Exploratory Data Analysis techniques using Python and provides valuable insights into passenger demographics, travel conditions, and survival outcomes.